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Young adults' desired life tradeoffs: love first, sex last

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Every human and non-human animal must make tradeoffs in investments in terms of time, energy, and resources. The aim of this study was to extrapolate from the types of investments in survival and reproduction that non-human animals make and translate these into human motivations. 16 potential goals were presented to 851 childless, 18–23-year-old adults from 11 world regions in an online study. Each young adult was asked to weight the importance of every goal to his or her ideal life. Weights had to sum to 100, requiring tradeoffs. Results revealed striking agreement across young adults with only four goals weighted above chance: Finding a beloved romantic partner, being physically and emotionally healthy, and earning money or resources. Having lots of sexual partners was the least important goal across all world regions for both sexes. Nevertheless, men more than women valued having many sexual partners, being talented outside work, being physically strong, and having a physically attractive romantic partner. Overall, there was cultural variation in some of the less important goals. Helping young adults achieve success requires understanding their own goals, rather than focusing on popularized depictions of what young adults desire.

Keywords Goals, Love, Motivation, Priorities, Sex, Young adults

Understanding young adults' goals should be critical to improving the well-being of both individuals and society. This is critical as recent reports document the major mental health difficulties young women and men are currently experiencing in the United States and other developed countries[1–3]. Our goal was to examine young adults' basic survival and reproductive goals.

Highly influential theoretical frameworks have been generated to explain humans' goals or motivations, such as McClelland's needs for achievement, affiliation, and power[4] or Maslow's hierarchy of needs from the physiological to self-actualization[5]. Important inferences about human motivations also arise from hundreds of behaviors that occur consistently across cultures[6,7]. Still others have focused on the overriding importance of specific motivations, such as the desire to belong to a social group[8] or to have an attachment figure as a child[9].

Although Maslow considered non-human animals in constructing his hierarchy of human needs, he pointedly did not believe that knowledge of non-human animals would contribute to understanding most human motivations[5]. In marked contrast, an evolutionary basis for human motivational systems predicated on natural selection has provided fruitful information about specific motives[10–12]. Furthermore, some evolutionary theories have been built specifically upon Maslow's original hierarchy of needs, such as by extending human motivations to include mating and parenting[12].

Life history theory is one form of evolutionary theory that focuses on the tradeoffs that individuals must make in allocating energy to survival, growth, and reproduction[13]. Attainment of an organism's life history goals requires balancing and weighting differing goals in order to maximize fitness.

The purpose of this study was to employ life history theory to examine the desired goals of young adults. Young adults were chosen because they are maximally fertile[14] but do not yet have children and in most developed countries have not yet settled into their adult lives. Our aim was to understand how much young adults weight the importance of goals that have been identified in non-human mammals as impacting survival and/or reproduction. To this end, we selected 16 personal and social goals that can be roughly divided into three categories: Reproduction, individual goals related to survival and/or reproduction, and alliances enhancing survival and/or reproduction. Universally documented sex differences in human mating predispositions[15] were also included to examine their importance relative to one another and to survival needs. Rather than indicating the importance of each goal, participants had to make trade-offs by prioritizing some goals over others. Our aims were to explore young adults' tradeoffs to determine whether there were any consistent cross-cultural priorities and whether sex differences exist in survival or reproductive goals.

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Reproduction

From an evolutionary perspective, the goal is to pass on genes to the next generation[16,17]. Although this can be accomplished indirectly, such as by helping close kin with reproduction, the most certain route is to bear one's own children[18]. The most important question therefore consisted of the *desire for children*. Whether this is a conscious motivation however is unclear. It may be that there is no need to desire children, as staying alive and finding a partner with whom to mate is sufficient to produce children. With the introduction and acceptance of effective birth control however, the desire for children should play a large role in whether an individual actually passes on his/her genes.

Individual goals that enhance survival and/or reproduction

Survival is the most basic motivation for animals, considered the basis of even Maslow's motivational hierarchy[5]. Survival also is related to more reproductive opportunities.

Survival is strongly enhanced by an individual's condition[19]. Specifically, animals that are healthier, bigger, stronger, tougher, and exhibit skills most adapted to the niches they occupy typically are more likely to survive and then reproduce, especially during periods of competition and environmental stress. Although the motivation to survive is rarely investigated in humans, it seems reasonable that humans too would be highly motivated to enhance their own survival and reproductive opportunities.

Survival may be more important for female than male mammals, because of different constraints on reproduction[20]. Compared with males, reproduction by females typically requires more extended care of offspring, while males may tradeoff survival for enhanced mating success[21]. Females' extended survival therefore should exert a greater impact on offsprings' survival, which is one probable explanation for why mammalian females typically live longer than males[22]. In humans likewise, evidence shows that across simple societies, mothers contribute more than fathers to offsprings' survival[23] and display more self-protective behaviors[24].

We chose seven basic goals that could enhance survival and reproductive success: Health, energy, physical strength, emotional strength, resource acquisition, skills at work, and additional talents. While we focus primarily on survival, many of these goals also would enhance reproductive success through either extending the time available for reproduction or enhancing mating or parental investment.

Health is the most evident proxy for and predictor of survival. In humans, self-reported health predicts survival[25]. *Energy* underlies all biological processes necessary for survival and/or reproduction with metabolic rate or daily energy expenditure major components of many animal studies[26]. Energy production in animals including humans typically requires tradeoffs between survival and reproduction[27]. In contrast, fatigue in humans signals reduced capacity to invest in all aspects of life[28].

Physical strength is also related to survival. Loss of physical strength is associated with mortality in non-humans such as wild and captive lemurs[29]. Likewise, reduced physical strength is a major predictor of mortality in older humans[30]. In younger individuals, physical strength plays a greater role in the mating success of mammalian males than females[19]. Further, across the animal kingdom males participate more than females in physical fights[31,32], including in humans[33]. *Emotional strength* has also been associated with survival[34]. As an example, a meta-analysis with non-humans shows that boldness often reduces survival in sexually immature individuals but enhances reproductive success in adults[35]. In humans, adult conscientiousness often predicts enhanced survival[36].

Survival is enhanced most directly by obtaining resources, including territories rich in food and safe from predators and adversarial conspecifics[19]. These resources also may be more important to mammalian females than males because of females' need to invest not only in themselves but also in their offspring over an extended period[32,37]. In humans, males who can provide resources then can enhance their mating success by attracting females[15]. In humans, socioeconomic status, which is measured by education, income, and occupation, and enhances resource acquisition, is strongly associated with greater survival[38,39]. *Earning money or resources* therefore is an important goal.

Specific skills aid in the acquisition of resources. Thus, skills such as hunting in chimpanzees[40] or the accumulated knowledge of aged female elephants[41] provide major survival and reproductive advantages to an individual. In humans, *work skills* as well as *outside talents* valued by the community enhance socioeconomic status which increases access to resources, again predicting enhanced survival[38,39]. Health, energy, physical strength, emotional health, earning money or resources, work skills, and outside talents all would be valuable not only for survival but also for attracting mates and investing in children.

Alliances that support survival and/or reproduction

Extensive research with mammals, including humans, demonstrates the benefits of many types of alliances to survival[42–44] and reproductive success for one or both sexes[15,45–47]. Because in many species the two sexes often differentially invest in parenting versus mating[20], based on prior research we hypothesized that females and males would differ in the types of alliances that were important to them[15,45]. Thus, we examined the importance of forming a pair bond; of being physically attractive to mates; of having a partner with resources or who is physically attractive; of forming alliances with kin, friends, and the community, and of having many sexual partners.

For pair-bonded species, one of the most important bonds is between mates regardless of whether monogamy is genetic or social[48]. About 10% of mammals form pair bonds[49], including humans for whom monogamy is the most common mating system[47]. Neural substrates, receptors, and hormones related to pair bonds have been identified, most prominently in prairie voles or monogamous mice[50,51]. Although mechanisms associated with pair bonds in humans are only beginning to be understood, pair bonds, ideally between two *mutually beloved romantic partners*, are present cross-culturally[47,52] and theorized to be a human universal[53].

Within every species, individuals differ in their *physical attractiveness* to potential mating partners[54]. For example, chimpanzee males find old, multiparous chimpanzee females most attractive as mating partners[55],

whereas female fallow deer find physically strong males, who can prevent other males from harassing them, most attractive[56]. For humans who typically form extended romantic/sexual partnerships, mate preferences also are present and influenced by physical attractiveness[15,45]. Whether women and men differ in how physically attractive they want to be themselves however has not been examined often or in relation to other priorities. Worldwide, women invest more heavily than men in beauty products (https://www.statista.com/topics/3137/cosmetics-industry), suggesting women may care more about being more physically attractive. This conclusion is supported by consistent findings that men more than women want their romantic partners to be physically attractive[15,45,46]. Nonetheless, across species, including humans, males typically confront more difficulty than females finding mates[57]. This suggests that it is men who should be more concerned than women about their attractiveness to mating partners. This potential contradiction may be resolved by a small study that showed no sex differences in desire for personal attractiveness[58]. Because personal physical attractiveness should theoretically be important to the formation of romantic partnerships, we include it as a potential goal.

Specific types of alliances however may be more beneficial to one sex than the other. For example, across the animal kingdom, alliances with another individual who shares food can be life-saving such as between female vampire bats[59], African wild dogs[60], or bonobos[61]. Further, paternal investment occurs in about 5% of mammals[62]. In humans, husbands' provisioning of wives, particularly when women are pregnant and lactating, is considered a major benefit of pair bonds[63–65] but see[66]. Cross-cultural research with humans consistently indicates that *receiving money or resources from a romantic partner* may be more critical to reproductive success of women than men[45,46].

In turn, as previously described, across cultures men more than women prefer *romantic partners who are physically attractive*[15,45,46], which usually means the women are young and fertile. This preference likely occurs because older women undergo menopause and can no longer reproduce, making them less desirable as mating partners. In contrast, men who typically are most attractive to women are older and more experienced and consequently often better able to provide resources and protection[15,45,46]. For some men who have accumulated resources, lack of physical attractiveness may be compensated for by their resources.

The importance of types of alliances also may differ by sex. Females may benefit more than males from *alliances with kin* in terms of enhancing their offsprings' survival, whereas males may benefit more than females from *alliances with friends* or non-kin in terms of gaining access to mates[67]. This is a compelling explanation for why mammalian females are more likely than males to remain in their natal communities their whole lives[19]. For human females who typically do not remain in their natal communities their whole lives however, when they give birth and have young infants, they often choose to live near their own mothers who provide valuable support[68]. Proximity to a grandmother who can invest in her grandchildren is associated with increased survival of grandchildren[23,69]. Likewise, some evidence suggests that human males benefit more than females from cooperation with unrelated same-sex peers[70–72].

Forming ties to the community is another important type of alliance. Structural and functional *social ties to the community* enhance longevity for both sexes[42–44].

Finally, we include the goal of finding multiple mating partners. Sex differences in the benefits of additional mating partners have been the subject of intense study across animal species[57,73,74]. Polygyny is the most common mammalian mating arrangement, indicating the large advantage to males of monopolizing several mating partners[21,75], which occurs in humans across the world as well[47]. More recent research however indicates that polyandry provides benefits to females, including humans[74]. Nonetheless, it is well-established that cross-culturally compared to women, men state that they desire a greater *number of sexual partners*[76,77]. In sum, being physically attractive, having a romantic partner who is physically attractive, receiving money or resources from a romantic partner, having a physically attractive romantic partner, forming alliances with kin, friends, or the community, and having many sexual partners are important factors that likely enhance both survival, and particularly, reproductive success.

The purpose of this study was to investigate young adults' priorities by requiring them to make explicit tradeoffs among these 16 goals (see Table S1). We chose to include childless, young human adults, ages 18–23 years, because they are sexually mature but still have many potential life history decisions to make. For contemporary humans, based on data from simple societies, women's maximal child-bearing age peaks in their early 20s[78]. Likewise, for men testosterone peaks in their late teens and early 20s then slowly declines with possible temporary increases[79]. Males typically produce offspring later than females in both non-humans[19] and humans[80]. Nonetheless, given hormonal changes in both sexes beginning in puberty, young adult humans who have recently become sexually mature are highly motivated to mate in their late teens and early 20s. At the same time, young adults have not yet realized many goals so that they are free to at least think about their ideal lives without the constraints that arise from having already chosen a specific life trajectory.

We conducted this study with two aims. First and foremost, using our tradeoff model we examined whether young adults from around the world exhibit any consistent survival and reproductive priorities. Second, we asked whether consistent sex differences exist in any survival and reproductive goals with a particular focus on sex differences that have been well-documented across human cultures.

Results

Consistency across regions

To determine whether there was any consistency across regions in priorities, we first calculated regional means for each of the 16 goals. We then computed an intraclass correlation across the 11 regions. This produced an $\alpha = 0.990$. As an additional measure of consistency, we calculated the intraclass correlation across all 851 participants, which produced an $\alpha = 0.997$. The striking consistency in young adults' priorities is depicted in Table 1 in which goals are ranked ordinally by the average weight they received by region (see Table 1). Because of the

Goal	Mean (SD) Weight n = 851	Canada USA n = 71	Chile n = 44	Eastern Europe n = 132	Japan Korea n = 9	Mexico n = 92	Oceania n = 31	Scandinavia n = 27	South Africa n = 83	Southern Europe n = 200	UK n = 48	Western Europe n = 114
Mutually beloved romantic partner (d = 0.84)[a]	17.73 (13.63)	1	1	1	3	2	2	1	3	2	1	1
Healthy (d = 0.79)[a]	17.33 (14.02)	2	3	2	2	1	1	2	2	1	2	2
Earn money or resources (d = 0.50)[a]	12.65 (12.75)	3	2	3	1	3	3	3	1	3	3	3
Emotionally strong (d = 0.12)[a]	7.16 (7.54)	4	5	5	4	5	6	10	4	4	5	5
Skilled at work (d = 0.08)	6.88 (7.98)	5	4	4	7	4	5	9	5	5	7	6
Energetic (d = − 0.07)	5.82 (6.46)	7	6	6	11	6	4	7	8	6	6	4
Talent outside work (d = − 0.22)[b]	4.82 (6.48)	9	7	8	6	7	8	4	11	8	9	8
Physically attractive (d = − 0.32)[b]	4.38 (5.80)	6	8	7	13	8	10	5	7	10	8	12
≥1 Children (d = − 0.29)[b]	4.20 (7.06)	12	11	10	8	13	13	12	6	7	4	10
Close family nearby (d = − 0.32)[b]	4.16 (6.47)	8	10	9	14	11	9	8	13	9	10	7
Good relationship with community members (d = − 0.38)[b]	4.09 (5.71)	11	9	11	9	9	7	6	10	11	11	9
Physically strong (d = − 0.72)[b]	2.93 (4.61)	13	14	12	5	10	14	15	12	12	12	14
Close friends nearby (d = − 0.72)[b]	2.77 (4.83)	10	12	14	10	15	11	11	15	13	13	11
Physically attractive romantic partner (d = − 0.77)[b]	2.61 (4.74)	15	13	13	15	12	12	13	14	14	14	13
Romantic partner earns money or resources (d = − 1.05)[b]	1.87 (4.16)	14	15	15	12	14	15	14	9	15	15	15
Many sexual partners (d = − 1.98)[b]	0.60 (2.86)	16	16	16	16	16	16	16	16	16	16	16

Table 1. The 16 goals ranked by overall mean weighting compared to chance ([a]above and [b]below) plus rankings by region.

high level of consistency and the large differences in sample sizes between regions, subsequent analyses were conducted with the combined full sample.

Priorities versus unimportant goals

We then examined which goals were most highly prioritized. To this end, we conducted a two-tailed, single-sample t-test to compare the average weight of each goal to 6.25, the expected weight if goals were selected randomly. Because we conducted 16 tests, we applied a Bonferroni correction, with the significance level set below 0.05/16. Table 1 displays the mean weights across all participants and indicates whether each weight was significantly higher or lower than chance. As can be seen from the table, only four goals were weighted significantly higher than would be expected by chance. We label these young adults' *priorities*. These were, in order,

having a mutually beloved romantic partner, being healthy, earning money or resources, and being emotionally strong. These four goals composed at least 50% of the sum of the weights used by young adults in 10 of 11 regions. Being skilled at work and being energetic did not differ from chance in their weights. The remaining ten goals were weighted significantly lower than what would be expected by chance. Although we cannot distinguish simply unimportant from potentially aversive goals, for simplicity we label them *unimportant*. Goals evaluated as unimportant were having a talent outside work, being physically attractive, having one or more children, having close family nearby, having a good relationship with community members, being physically strong, having close friends nearby, having a physically attractive romantic partner, having a romantic partner who earns money or resources, and having many sexual partners. The least important goal overall, many sexual partners, was ranked least important in all 11 of the regions.

We then compared the relative importance of the four priorities in more detail. Table 2 provides the weights by region for the four priorities (see Table 2). Examination of the distribution of these weights showed that they were positively skewed, so we applied a square root transformation. Then, the four square-root transformed weights were entered into a repeated measures linear mixed model in R 4.3.2 with sex as an independent variable and participant nested within region. Table 3 displays the results which showed a significant effect only of priorities (see Table 4). Contrasts using Tukey's test show that having a mutually beloved romantic partner and being healthy were young adult's highest priorities for an ideal life. These two top priorities did not differ from one another in weight, $p = 0.999$, but were significantly more important, $p < 0.0001$, than earning money or resources which, in turn, was significantly more important than emotional strength, $p < 0.0001$.

Sex differences in the importance of goals

Finally, sex differences were individually examined for all 16 goals. Due to the positive skew of the goals, two-tailed Mann–Whitney U tests with Bonferroni corrections, 0.05/16, were conducted to compare young women versus men in the weight they attributed to each goal. Table 4 displays the results (see Table 4). Individually, none of the four priorities exhibited significant differences between women and men. Instead, only four unimportant goals exhibited sex differences. Men more than women wanted to exhibit a talent outside of work and to be physically strong. Furthermore, consistent with past literature, men more than women also wanted to have a physically attractive romantic partner and many sexual partners. However, the effect sizes for all sex differences were small.

Goal	Canada/USA (n = 71)	Chile (n = 44)	Eastern Europe (n = 132)	Japan/Korea (n = 9)	Mexico (n = 92)	Oceania (n = 31)	Scandinavia (n = 27)	South Africa (n = 83)	Southern Europe (n = 200)	UK (n = 48)	Western Europe (n = 114)
Mutually beloved romantic partner	16.94	16.41	18.71	15.11	15.37	15.07	21.11	11.84	19.14	20.52	20.28
	(13.01)	(10.11)	(14.07)	(15.33)	(11.26)	(9.73)	(15.65)	(11.80)	(14.42)	(13.94)	(15.00)
Healthy	15.41	14.39	16.86	16.33	17.14	15.29	13.74	14.27	20.00	17.02	19.54
	(13.99)	(11.41)	(13.09)	(14.44)	(13.27)	(9.04)	(10.46)	(15.18)	(15.34)	(11.56)	(15.32)
Earn own money or resources	12.27	15.32	13.27	16.67	12.42	10.36	10.82	21.36	10.91	10.58	9.61
	(10.44)	(12.23)	(12.55)	(13.92)	(9.92)	(12.74)	(9.45)	(18.45)	(12.28)	(11.54)	(10.60)
Emotionally strong	7.59	7.89	6.32	10.44	8.34	7.23	5.11	7.24	7.42	6.52	6.58
	(6.76)	(7.99)	(6.91)	(10.15)	(7.74)	(6.37)	(5.40)	(7.73)	(7.83)	(9.09)	(7.51)
Total mean weight of four priorities	52.21	54.00	55.15	58.56	53.27	47.94	50.78	54.71	57.47	54.65	56.02
	(17.07)	(19.63)	(18.39)	(22.86)	(17.92)	(18.60)	(16.13)	(20.77)	(18.30)	(17.25)	(18.24)

Table 2. Mean (SD) weights of four priorities by region.

Priority	Estimated marginal means	SE	Lower confidence level	Higher confidence level
Mutually beloved romantic partner	4.18	0.275	3.64	4.71
Healthy	4.14	0.275	3.60	4.68
Earn money or resources	3.45	0.277	2.90	3.99
Emotionally strong	2.39	0.275	1.84	2.95
	Chi-square	**df**	***p***	
Priority	490.42	3	< 2e−16	
Sex	0.10	1	0.751	
Priority × Sex	5.76	3	0.124	

Table 3. Results of linear mixed model on priorities with sex as an independent variable.

Attribute	Females Mean (SD) (n = 412)	Males Mean (SD) (n = 439)	Mann–Whitney U[a]	*p*	d
Mutually beloved romantic partner	18.73	16.79	82,825	0.032	− 0.07
	(13.90)	(13.31)			
Healthy	17.60	17.09	87,686	0.438	− 0.03
	(14.11)	(13.95)			
Earn money or resources	12.86	12.44	90,181	0.943	− 0.00
	(13.11)	(12.41)			
Emotionally strong	8.02	6.35	81,979	0.015	− 0.08
	(8.40)	(6.55)			
Skilled at work	6.56	7.18	94,236	0.274	− 0.04
	(7.28)	(8.57)			
Energetic	5.60	6.04	96,987	0.058	− 0.06
	(6.85)	(6.08)			
Talent outside work	4.20	5.41	111,919	3.962e−10	− 0.21
	(6.22)	(6.68)			
Physically attractive	4.35	4.41	92,842.5	0.477	− 0.02
	(5.99)	(5.62)			
≥1 Children	4.12	4.27	98,044	0.019	− 0.08
	(7.50)	(6.63)			
Close family nearby	4.68	3.67	84,884	0.093	− 0.06
	(7.01)	(5.88)			
Good relationship with community members	4.23	3.96	89,846	0.862	− 0.01
	(6.05)	(5.37)			
Physically strong	2.02	3.79	108,117	4.154e−08	− 0.19
	(3.50)	(5.31)			
Close friends nearby	2.97	2.58	88,534	0.547	− 0.02
	(4.99)	(4.67)			
Physically attractive romantic partner	1.51	3.65	111,400.5	2.12e-11	− 0.23
	(3.03)	(5.72)			
Romantic partner earns money or resources	2.31	1.44	83,321.5	0.013	− 0.08
	(4.68)	(3.56)			
Many sexual partners	0.23	0.94	100,986	3.54e-07	− 0.17
	(1.14)	(3.79)			

Table 4. Sex differences in the importance of goals. [a]Mann–Whitney U tests were used to compare ordinal ranks for females and males for each goal. Goals in bold exhibit significant sex differences following a Bonferonni correction for 16 comparisons.

Discussion

In this self-report study, we asked young adults from multiple world regions to indicate how they would tradeoff goals related to survival and reproduction. A remarkably consistent profile emerged in which a beloved romantic partner and personal health were the two most important goals across all regions followed by having money or resources. Emotional strength was also prioritized, but less so. This study therefore provides insight into young adults' goals which could guide those who are attempting to provide support to this age group.

Having sex with multiple partners for these young adults was least important for both young women and men across all world regions. This finding suggests that popular media and research accounts that highlight the importance of sexuality to young adults may not reflect young adults' greatest concerns and instead provide a distraction.

Besides having sex with multiple partners, 11 other goals were weighted as relatively unimportant. Surprisingly, these included having children, being physically attractive, being physically strong, being skilled at work, and having good relationships with others besides a romantic partner. Again, these goals typically form major parts of the public discourse and research investigations, yet young adults at least do not seem to value them as much as they do love, health, and resources.

Four sex differences in the importance of the goals emerged. Compared to women, men weighted more heavily having a talent outside work, being physically strong, having a physically attractive romantic partner and having many sexual partners. The latter two have been reported consistently across cultures[45,46], providing validity for the current study. Being physically strong, having an attractive (fertile) romantic partner, and having many sexual partners likely play a greater role in mating success for men than women. The reason having a special talent is more important to men than women is unknown, but it may also be linked with an additional

means of acquiring resources. Unexpectedly however, none of these goals was prioritized among young adults. All together, these results de-emphasize the importance of sex-differentiated goals in young adults' lives.

Interestingly, we did not find any sex differences in the importance of earning money or resources or being physically attractive. Both women and men benefit from having resources and from being desired as a mate to the same degree. Although women more than men did prefer a romantic partner who earned resources as found in prior research[15], the sex difference was not large enough to attain significance in this study. Finally, despite past findings, women and men did not differ in the perceived benefits of having nearby kin or friends, or good relationships with community members. Moreover, none of these bonds were prioritized when tradeoffs were required.

Critically, the 16 goals chosen for this study were based on personal and social attributes that have analogs in non-human animals. Nonetheless, they are not exhaustive. Because the current results were based on tradeoffs, the findings are strongly affected by which goals were included. To assess the validity of our goals, we asked participants to describe any missing goals. All responses are provided (see Table S2). When we examined additional goals provided by the participants, they varied greatly from the highly personal to the abstract. Therefore, there was no consensual missing goal, providing some validity for our choices. Importantly, few of the additional goals were directly related to survival or reproduction, and many overlapped with the 16 goals we provided. Nonetheless, the most important goals that affected survival and reproductive success that were provided were related to physical environmental attributes, which we did not assess. These included living in a safe neighborhood, having access to clean air and water, and earning money and resources in a satisfactory environment, all of which would be critically important to non-human animals as well.

A limitation is that half of our sample came from Europe, which may have inflated the consistency of the responses. We specifically targeted Europe because of its multiple diverse countries, cultures and languages compared with North America. Had more countries been available online from other continents, however, this would have provided greater support for our results.

Our approach was predicated on humans' sharing fundamental life goals with non-human animals. Nonetheless, species and even populations within species differ markedly in their social, cognitive, and physical adaptations, as well in the physical and social settings they inhabit. Differing social organizations, mating practices, and caring systems modify goals[81]. We expect therefore that priorities differ depending on species and populations as well as changing environments.

In humans in particular, cultures differ along multiple dimensions[82,83]. Numerous human motivations have been investigated across differing cultures, cohorts, and ages[82,84–88], including goals in young children or ones that are not easily accessible to human consciousness and consequently cannot be measured with self-reports[89]. Additionally, individual differences in priorities are common within a culture[84,88]. Further, goals and priorities can change within populations and across generations[85], and even over short periods of time, such as when pathogens suddenly appear[90]. Thus, in our study, the current prioritization of emotional health may be particularly important to the cohort and time interval we included, as compelling cross-cultural research suggests[2,3,91]. Ultimately, specific priorities depend on what is life-threatening or threatening to reproductive success to individuals at a particular time in a specific context.

At the same time, we suggest that satisfying basic life-affirming and reproductive goals always will be priorities and can provide a strong foundation for understanding motivation, one that often has been neglected. Accordingly, the desire for health and resources likely would be priorities across many species, populations, individuals, and contexts over time. Additionally, in humans, the desire to form pair bonds likely has an underlying biological basis that supersedes many other goals[53,92]. Whether this desire occurs in the few polygynous human societies that exist is unknown but research suggests that it may be present even in polygynous communities[93].

Recent research has identified a life stage labelled "emerging adulthood," which includes the ages of the participants in our study[87,94,95]. Emerging adulthood typically is characterized as a time of identity exploration, instability, self-focus, feeling "in-between," and possibilities. It includes variables that focus on personality, politics, religion, gender and sexuality, and ethnicity[86]. Overall, emerging adulthood is a stage believed to accentuate changing goals and individual differences. Our study provides a different perspective. When tradeoffs involving basic survival and reproductive goals are required, the priorities listed by the young adults in our study were highly consistent. Nonetheless, we concur that priorities would differ in older adults, particularly once life trajectories have narrowed and fewer options are available.

Other limitations of our study include its dependence on self-reports which may not reflect actual behavior. The goal of this study, however, was to examine young adults' ideal goals, unimpeded by the difficulties of actually attaining those goals. Consequently, self-reports provided valuable information as they likely assess behavioral motivations.

Another shortcoming of this study is that the participants were limited to young adults who could understand English and use computers, some of whom, but not the majority, had completed university. An older or less educated sample or from a less developed nation sample might make different tradeoffs. Nevertheless, the current study provides some insight into what a portion of the world's young adult population desires in their lives at this point in time.

Strikingly, the desire to have children was weighted as being relatively unimportant. From an evolutionary perspective, the desire for children should constitute the most important priority for both sexes. Although this goal's weighting was particularly variable across regions, if these weights translate into behavior, then most participants would be expected to have relatively few or no children. This suggests a novel explanation for the demographic transition to reduced fertility. Rather than fertility declines resulting solely from benefits derived from investing more heavily in fewer offspring, transmission of material wealth, or increased lifespan as has been theorized[96], it may be that the desire to have children is not an evolved motivation. Instead, evolution may have relied on the desires to form romantic/sexual bonds to produce children.

It is only where effective birth control practices are practiced that the lack of desire for children would exert its effect. The lack of desire for children is consistent with greater female empowerment being related to reduced fertility[97]. That is, when women have choices, they do not necessarily desire to produce children. Because we excluded young adults who already had children from participating, it is possible that our results are underweighting the importance of the desire to have children. At the least, the large variation in fertility rates in the last few generations and as recently reported across diverse populations[98] combined with widely differing importance of having children reported across regions in this study suggests that the desire to have children is highly conditional on local conditions and not uniformly strong. While it is possible that as they age, these young adults will feel more inclined to reproduce, both women and men become less fertile as they age[14]. This would reduce their chances of ever reproducing.

In summary, in studies on motivation, it is rare to focus on the basic survival and reproductive goals that affect all living individuals. We suggest that a life history approach may provide a more grounded and consensually based means of understanding the importance of basic human goals. Practically, the present results suggest that understanding and addressing young adults' priorities might contribute to addressing some of the current factors underlying a surge in young adults' emotional and mental health difficulties.

Method

The survey was conducted on the Qualtrics platform https://login.qualtrics.com/login?lang=en.

Participants were recruited through the online research platform Prolific https://www.prolific.com/ and were paid 1£ for participating in a 4-min study. Informed consent was obtained from all participants.

Participants

Only individuals who self-identified as childless women or men between 18 and 23 years inclusive were included. World regions were based upon geographic proximity. Within each world region, countries that included at least 25 women and 25 men who met the selection criteria were targeted. We then advertised our study to all individuals within those countries that met the selection criteria during June of 2023. The number of individuals accepting our invitation to participate varied greatly by country and region Rather than omitting any participants from countries/regions with fewer than 50 participants, we included every participant who satisfied our selection criteria.

The final sample consisted of 851 participants (412 women, 439 men) from 30 countries. Participants in countries from the same geographical area were grouped into 11 regions, which varied widely in size from 9 to 200 participants: Canada/USA: Canada (29), United States of America (42); Chile (44); Eastern Europe: Czech Republic (10), Estonia (15), Hungary (41), Latvia (18), Poland (27), and Slovenia (21); Japan/Korea: Japan (6) and South Korea (3); Mexico (92); Oceania: Australia (18) and New Zealand (13); Scandinavia: Denmark (4), Finland (9), and Sweden (14); South Africa (83); Southern Europe: Greece (89), Israel (29), Italy (4), Portugal (31), and Spain (47); United Kingdom (UK) (48); Western Europe: Austria (1), Belgium (14), France (24), Germany (30), The Netherlands (44), and Switzerland (1). After initial recruitment, for those regions in which fewer than 20 women and 20 men were recruited, extra efforts were made to recruit additional individuals. Greece was inadvertently recruited twice.

By definition, all participants had access to Prolific on a computer and understood written English. Information on education was not collected for 198 participants (98 females) primarily from Chile, Mexico, and South Africa. For the remainder, 112 had a high school education or less (44 females, 68 males), 355 had some university or technical training (168 females, 187 males), and 186 had graduated from university (102 females, 84 males).

Procedure

All consenting participants were presented with instructions to select their priorities from a list of 16 (see Table S1). However, tradeoffs were required as participants had to weight the relative importance of each goal, ensuring that the total weights summed to 100. After participants made their selections, they were asked whether there were any priorities that had not been listed. Finally, participants were thanked and paid £1 through Prolific.

To check whether we had missed important goals, we examined any additional goals that participants provided. We provide the complete list (see Table S2). Fewer than half of the participants provided additional goals, and there was no clear consensus regarding any missing goals.

Ethical approval

The study was approved by the Research Ethics Committee of the University of Quebec at Montreal. The methods used were performed in accordance with relevant guidelines.

Data availability

All data are available at https://osf.io/2kzw9/.

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Author contributions

J.B. was primarily responsible for conception, data collection, methodology, and statistical analyses. J.B. wrote the first draft. H.M. contributed to the methodology and statistics and revised the manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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